Margaret Lee

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EDUCATION

University of British Columbia | Vancouver, Canada *05/2026*
Engineering Physics, BASc
CGPA: 3.95, Awards: Roy Nodwell Memorial Prize (2025), Dean's Honour List (2020–2025), Trek Excellence Scholarship (2022)

ETH Zürich | Zürich, Switzerland *09/2023 – 02/2024*
Mechanical Engineering, International Exchange
Courses: Robot Dynamics, Microrobotics, Space Research and Exploration, Virtual Reality, Distinguished Seminar in Robotics

WORK EXPERIENCE

Red Rabbit Robotics | Vancouver, Canada *01/2026 – 04/2026*
Hardware Engineering Intern
- Designed and iterated precision grippers for a humanoid robot, prototyping via 3D printing and in-house CNC machining with integrated compliant elements for reliable manipulation of unstructured material
- Owned the mechanical–software interface by generating URDF models from CAD for ROS simulation, working cross-functionally to validate kinematics, joint limits, and integration requirements prior to hardware build
- Evaluated manufacturing pathways (in-house, local, overseas) and developed engineering drawings for suppliers

Formlabs | Somerville, MA, USA *09/2025 – 12/2025*
Hardware R&D Print Process Intern
- Reduced per-layer print time by 15% by developing time-optimized Z-stage motion trajectories through analysis of 2,000+ trials with Python, maintaining mechanical stability and print quality
- Analyzed force, temperature, and ultrasonic sensor data to diagnose layer-level failures on next-gen SLA printers, mapping sensor feedback to motor positions to identify mechanical drift and force anomalies
- Characterized debris flow dynamics with OpenCV-based particle tracking to refine Z-stage control parameters

Cyber-Physical Systems Group, University of Konstanz | Konstanz, Germany *05/2025 – 08/2025*
Swarm Robotics Researcher (DAAD RISE Scholar)
- Developed autonomous sailboats to contribute to swarm control and limnological research on Lake Constance
- Implemented real-time obstacle avoidance using multimodal sensor fusion (mmWave radar, RGB and thermal cameras) and classical computer vision, integrating sensing hardware with onboard processing on Raspberry Pi

Microchip Technology Inc. | Burnaby, Canada *05/2023 – 08/2023*
Product Engineer Co-op
- Designed automated test flows with Python, thermal forcers, and Excel to characterize temperature sensors
- Identified calibration hardware resolution limits causing temperature sensor errors; developed a software workaround that recovered full sensor accuracy and prevented multi-month shipment delays for 1,000+ chips

TECHNICAL PROJECTS

UBC Sailbot Design Team *11/2021 – Present*
Polaris, Mechanical Team
- Led mechanical design and manufacturing of the hull for an autonomous sailboat, directing subsystem teams and integrating mechanical, electrical, and software requirements under performance and stability constraints
- Designed hull, internal structures, and sensor mounts in SolidWorks; standardized collaboration workflows for team-wide 700+ part assembly
- Optimized hydrostatic, hydrodynamic, and structural properties using ANSYS FEA and Maxsurf simulations

UBC Engineering Physics Project Lab *09/2024 – 04/2025*
LoRa Pet Tracker
- Designed a wearable PCB integrating STM32 MCU, LoRa, GPS, and IMU for real-time pet location monitoring
- Built custom C libraries for sensor drivers and integration over I2C, SPI, and UART, optimizing for power and range
- Recipient of the Roy Nodwell Memorial Prize for a high professional standard and industry relevance

UBC Psychiatry NINET Lab *09/2023 – 04/2024*
TMS Cobot System
- Developed a 6-DOF motion compensation system using a UR3e cobot, ROS Noetic on Linux, and OptiTrack IR tracking to maintain precise TMS coil alignment with the patient's head during treatment
- Integrated force/torque sensing for compliant control and safe human-robot interaction
- Implemented spatial transformations (tf2) and inverse kinematics for real-time pose correction with optical tracking

TECHNICAL SKILLS

Mechanical	SolidWorks, ANSYS, Maxsurf, CFRP composites, 3D printing, CNC, Milling, Lathe
Electrical	Oscilloscope, Signal generator, Soldering, Altium, PCB design and testing
Embedded	C, STM32, Firmware development, I2C, SPI, UART, Hardware debugging, Sensor integration
Software	Python (pandas, SciPy, OpenCV, NumPy, TensorFlow/Keras), MATLAB, ROS, Linux, Git